

FIT5145 Foundations of data science

[Rubric]

Assessment 1: Business & data case study - Proposal

Assessment 3: Business & data case study - Report

Assessment 1: Proposal (Rubric)

Criteria	Allocated percentage mark (Total: 15%)	High distinction	Distinction	Credit	Pass	Fail
Problem Clarity: Is the problem well-articulated and clearly defined? Is the goal of the project clearly stated?	2	Provides a sophisticated description of the project, including the problem and goals.	Provides some description of the project, including the problem and goals.	Provides a limited description of the project, including the problem and goals.	Provides a minimal description of the project, including the problem and goals.	Provides no clear description or a very insufficient description of the project, including the problem and goals.
Problem Importance: Does the project have real-world applications? Does it address key social, environmental, or business challenges and demonstrate potential for significant social impact?	4	Provides a sophisticated description of the problem importance by discussing the real-world application and the social impact.	Provides some description of the problem importance by discussing the real-world application and the social impact.	Provides a limited description of the problem importance by discussing the real-world application and the social impact.	Provides a minimal description of the problem importance by discussing the real-world application and the social impact.	Provides no clear description or a very insufficient description of the problem importance by discussing the real-world application and the social impact.
Business Model Analysis: Is the data business model of the project clearly articulated? Is the level of business analytics provided by the project clearly identified, such as descriptive, predictive, or prescriptive analytics? Are the benefits, value and	4	Provides a sophisticated description of the business model, including the data business model, analytical level, benefits/value/challenges	Provides some description of the business model, including the data business model, analytical level, benefits/valu	Provides a limited description of the business model, including the data business model, analytical level, benefits/valu	Provides a minimal description of the business model, including the data business model, analytical level, benefits/value/challenges	Provides no clear description or a very insufficient description of the business model, including the data business model, analytical level, benefits/value/challenges and stakeholders.

challenges of the project clearly outlined? Are the primary stakeholders identified and their needs addressed?		and stakeholders.	e/challenges and stakeholders.	e/challenges and stakeholders.	and stakeholders.	
Criteria	Allocated percentage mark (Total: 15%)	High distinction	Distinction	Credit	Pass	Fail
Novelty: Does the project address an important and novel problem? Does it introduce a new or unconventional approach? Does it tackle an underexplored or emerging issue in data science?	5	The project addresses a highly novel problem, introducing a unique or unconventional approach with significant potential impact.	The project addresses a relevant and novel problem, offering an original approach with clear contributions to the field.	The project addresses an important problem but with a conventional approach, showing moderate originality and creativity.	The project addresses a problem with a conventional approach and limited originality, meeting basic expectations.	The project lacks novelty, using a conventional or simplistic approach without original contributions.

Assessment 3: Report (Rubric)

Criteria	Allocated percentage marks (Total: 30%)	High distinction	Distinction	Credit	Pass	Fail
Introduction, Related Work and Business Model: define the problem, provide background and significance, outline specific goals, analyse the data business model and its value creation, identify key stakeholders and their benefits, summarize existing research or industry solutions, highlight gaps in current approaches, and justify the project's novelty and potential impact.	1	Provides a sophisticated description of the project, including the problem, goals, data business model, values, stakeholders, gaps, and novelty.	Provides some description of the project, goals, data business model, values, stakeholders, gaps, and novelty.	Provides a limited description of the project, goals, data business model, values, stakeholders, gaps, and novelty..	Provides a minimal description of the project, goals, data business model, values, stakeholders, gaps, and novelty.	Provides no clear description or a very insufficient description of the project, goals, data business model, values, stakeholders, gaps, and novelty.
Characterising and Analysing data: discuss potential data sources and analyse their characteristics (e.g., the 4 V's) and evaluate the required platforms, software, and tools for data processing and storage based on the specific characteristics of the data and consider potential options (e.g., platforms, software, and tools) if your project expands in the future.	3	Provides clear demonstration of the sources, characteristics of data, and data processing and data storage, and classification of the technologies in use.	Provides some demonstration of the sources, characteristics of data, and data processing and data storage, and classification of the technologies in use.	Provides limited demonstration of the sources, characteristics of data, and data processing and data storage, and classification of the technologies in use.	Provides minimal demonstration of the sources, characteristics of data, and data processing and data storage, and classification of the technologies in use.	No clear demonstration of the sources, characteristics of data, and data processing and data storage, and classification of the technologies in use.

Criteria	Allocated percentage mark (Total: 30%)	High distinction	Distinction	Credit	Pass	Fail
Characterising and Analysing data: specify the data analysis approach, including the overall process and the technical and statistical methods (e.g., machine learning techniques) applicable to the project. Provide a rationale for the selected approach and methods, and discuss the expected outcomes.	2	Provides a clear description of the data analysis approach, including the overall process and the technical and statistical methods.	Provides some description of the data analysis approach, including the overall process and the technical and statistical methods.	Provides a limited description of the data analysis approach, including the overall process and the technical and statistical methods.	Provides a minimal description of the data analysis approach, including the overall process and the technical and statistical methods.	No clear description of the data analysis approach, including the overall process and the technical and statistical methods.
Demonstration: identify a usable dataset for the proposed project and perform some analysis on the identified dataset to demonstrate the feasibility of the project, using R (e.g., extracting insights from the dataset and building machine learning models), and report the analysis process and result in the demonstration section of a final report	4	Provides clear demonstration on the project by using usable data and analysing it.	Provides some demonstration on the project by using usable data and analysing it.	Provides limited demonstration on the project by using usable data and analysing it.	Provides minimal demonstration on the project by using usable data and analysing it.	No clear demonstration on the project by using usable data and analysing it.
Standard for Data Science Process, Data Governance and Management: describe any standard used in your data science process, and practices for data governance and management in the project, e.g., how to address key issues such as data accessibility, security, and confidentiality, as well as potential ethical concerns related to data usage.	1	Provides clear description of standard, data governance and management	Provides some description of standard, data governance and management	Provides limited description of standard, data governance and management	Provides minimal description of standard, data governance and management	No clear description of standard, data governance and management

Criteria	Allocated percentage mark (Total: 30%)	High distinction	Distinction	Credit	Pass	Fail
Think critically and creatively, providing justification and analysis.	1	Thinks out of the box, creates or extends to a novel or unique idea. Provides a sophisticated critical analysis.	Collect ideas, solutions and other information in good ways. Provides detailed justification and analysis	Reformulates a collection of available information. Provide some justification and analysis.	Mostly repeats existing information. Provide limited justification and analysis.	Just repeats existing information. Do not provide any justification or analysis.
Provide a high-quality report in terms of structure, organisation, formatting, expression, grammar, spelling, referencing, and adherence to the word limit.	3	Excellent structure, organisation, and flow. Clear and professional writing with minimal errors. Consistent formatting and accurate referencing. Fully adheres to the word limit.	Well structured with good flow. Clear writing with minor errors. Appropriate formatting and referencing. Adheres to the word limit.	Satisfactory structure and flow. Generally clear writing with some errors. Mostly appropriate formatting and referencing. Generally adheres to the word limit.	Basic structure with limited flow. Writing contains noticeable errors. Some inconsistency in formatting or referencing. Loosely adheres to the word limit.	Poor structure and flow. Frequent writing errors affect readability. Inadequate formatting or referencing. Does not adhere to the word limit.
Recorded Presentation and Slide Quality: Provide a well-structured presentation with clear and natural delivery, effective slide content, appropriate language, and adherence to the required format and duration.	5	Excellent structure and flow, with clear, confident, and natural delivery. High-quality slides and full adherence to format and duration.	Well-structured with good flow and clear delivery. Good-quality slides and strong adherence to format and duration.	Satisfactory structure and flow with adequate delivery. Generally appropriate slides and reasonable adherence to format and duration.	Basic structure and flow with limited clarity in delivery. Weak slides and limited adherence to format or duration.	Poor structure and delivery, with unclear flow. Slides are ineffective and requirements are not adequately met.
Interview and Responses to Questions: Explain your project clearly and confidently in your own words. Justify your key decisions such as methodology, data, and techniques, showing why, not just what. Engage with follow-up questions by reasoning through trade-offs and alternatives, not repeating	10	Clear, confident command of the project, explained in their own words. Justifies key choices such as method, data, technique with sound reasoning. When probed, extends the reasoning: discusses	Explains the project accurately and justifies most choices. Handles probing adequately, but reasoning is underdeveloped on at least one point. Clearly understands the	Describes correctly what the project does, but justification is thin, mostly recalling the report rather than reasoning about it. Struggles when a follow-up goes beyond the surface; can state	Conveys a basic, partial understanding. Answers are vague, incomplete, or partly incorrect. Little justification; when probed, cannot extend or defend.	Cannot explain the project; answers largely incorrect, absent, or contradicting the report. No evidence of genuine understanding.

prepared answers.		trade-offs, limitations, or alternatives rather than repeating the report. Evident ownership of the work.	project; depth is good but not complete.	choices but not fully explain why. Understanding is real but shallow.	Noticeable gaps in understanding their own project.	
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